

Adaptive SIMEX for Poisson Regression with Classical Measurement Error: Theory, Methods, and Applications

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Abstract

Poisson regression is widely used for count outcomes, but covariates are frequently observed with classical measurement error, and the resulting naive estimates are attenuated with invalid confidence intervals. We develop Adaptive SIMEX (A-SIMEX), which automates the two ad-hoc choices of SIMEX, namely the extrapolation region and the polynomial degree, by cross-validation with a one-standard-error parsimony rule, adds a ridge-corrected second extrapolation stage, and is accompanied by complete asymptotic theory: consistency under a growing-degree regime that we make explicit; asymptotic normality with an exact decomposition of the asymptotic variance into data and simulation components (the latter vanishing at rate B^{-1} in the number of simulation replicates); and efficiency results relative to the oracle estimator, including an information-monotonicity lemma showing that no estimator based on noisy covariates can improve on the oracle to first order, while the second-order efficiency comparison is signed neither way. The theory corrects several claims of earlier SIMEX treatments and resolves the tension between consistency (which requires the polynomial degree to grow like $\log n$) and inference through an explicit window condition on the degree growth rate. In simulation studies with 1,000 Monte Carlo repetitions per design cell, including misspecified (Laplace) error, negative-binomial outcomes, error variance estimated from replicated measures, and corrected-score and regression-calibration competitors, A-SIMEX removes most of the naive attenuation with valid inference when the error is moderate, and we identify an important limitation of automatic tuning: held-out prediction on the observed scale systematically under-extrapolates when the error is large, because its population minimizer is the naive pseudo-parameter. We characterize the resulting bias–variance trade-off, quantify the efficiency premium of extrapolation against the oracle, and give practical remedies. The methods are implemented in an open-source R package with an analytic sandwich variance estimator that we validate against the parametric bootstrap, and are applied to NHANES 2017–2018, where the measurement error is directly measurable: self-reported weight in a model for the count of prescription medications. All results are reproducible from a documented archive with fixed seeds.

Keywords: measurement error, SIMEX, Poisson regression, asymptotic theory, generalized linear models, replication archive

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1. Introduction

Economists measure the covariates they regress on, and the measurements are frequently wrong. In labor and health economics, self-reported income, earnings, and weight differ from administrative records in well-documented ways, and the resulting demand and health production estimates inherit the discrepancy (Griliches, 1986; Bound et al., 2001). In demand systems estimated from household surveys, reported consumption shares are noisy versions of the expenditures that economic theory needs, so price and income elasticities are biased in unknown directions. In policy evaluation, treatment intensity and exposure are measured by program records or monitors that misreport systematically, misclassifying the very variable whose effect is of interest (Mahajan, 2006; Hu, 2008). Many of the resulting outcome variables, from doctor visits to patent counts to crime incidents, are counts, and the Poisson regression model with its overdispersed extensions is the standard tool for them (Mullahy, 1986; Cameron and Trivedi, 2013; Winkelmann, 2008). A long econometric literature documents that plugging mismeasured covariates into nonlinear models attenuates estimates, inflates variances, and destroys interval coverage (Hausman, 2001; Bound et al., 2001); Section 5.1 quantifies the damage cell by cell for the Poisson case.

In nonlinear models the problem is structurally different from the familiar regression-dilution algebra of the linear model. The pseudo-true parameter of the naive fit is defined only implicitly, through a nonlinear moment condition that mixes the error variance with the moment-generating function of X evaluated at the parameter itself; the attenuation is therefore not a simple constant and is not sign-monotone in all directions (Lemma 1(iii) computes it exactly to first order). Classical measurement error in GLMs is the setting of a substantial statistics literature (Carroll et al., 2006), and econometric work has developed semiparametric and identification-based alternatives (Lewbel, 1997; Schennach, 2004; Mahajan, 2006; Hu, 2008); yet for the applied count regression the practical menu remains short, and simulation-based corrections of the SIMEX type are far better known in biostatistics than in economics (Nakamura, 1990; Cook and Stefanski, 1994). This paper is written for the applied econometrician who has a count outcome, a mismeasured regressor, and either an external estimate of the error variance or replicated measurements, and who needs a correction that is automatic, documented, and honest about its own limits.

Simulation extrapolation (SIMEX; Cook and Stefanski, 1994; Stefanski and Cook, 1995) is arguably the most practical member of the correction toolkit: it makes no model for the distribution of X , requires only the error variance, reduces the correction to a sequence of ordinary GLM fits, and extends mechanically to misclassification (Küchenhoff et al., 2006) and semiparametric settings (Apanasovich et al., 2009). Yet SIMEX as usually practiced leaves the two consequential choices, the extrapolation region and the polynomial degree, to the analyst, offers no rigorous asymptotic theory for GLM settings, and provides no closed-form standard errors. Table 1 summarizes where the available approaches stand against the desiderata that motivate this paper: automatic tuning, formal asymptotics, usable inference machinery, freedom from validation data, and scalability to moderate dimensions. The table makes clear that no existing method occupies the full row that an applied econometrician needs; it also previews our claim that A-SIMEX fills the row, with one honest asterisk, namely that its automatic tuning inherits a documented under-extrapolation bias when the measurement error is large (Section 5.1).

This paper develops Adaptive SIMEX (A-SIMEX) and subjects it to a level of scrutiny that we argue should be standard for bias-correction methods. Our contributions are as follows.

1. **Automated hyperparameter selection, with its limits quantified.** Extrapolation region and polynomial degree are selected by k -fold cross-validation of the held-out Poisson

Table 1: Available approaches to Poisson (GLM) regression with classical covariate measurement error, against the desiderata motivating this paper. “Automatic” = no analyst-chosen smoothing/extrapolation tuning; “Inference” = analytic standard errors beyond point estimation. Superscripts flag caveats discussed in the text and quantified in Section 5.

	Automatic	Consistency theory	Inference machinery	No validation data required	Moderate- p scalable
Naive MLE (Fuller, 1987)	✓	^a	×	✓	✓
Regression calibration (Prentice, 1982; Rosner et al., 1989)	✓	^b	approximate	✓	✓
Corrected score (Nakamura, 1990)	✓	✓	✓	✓	^c
Deconvolution / functional (Fan, 1991; Delaigle and Meister, 2008)	×	✓	×	✓	×
Validation likelihood (Qin, 2000; Chen et al., 2011)	✓	✓	✓	×	✓
Bayesian hierarchical ME (Richardson and Gilks, 1993; Berry et al., 2002)	✓	^d	posterior	×	×
Standard SIMEX (Cook and Stefanski, 1994; Stefanski and Cook, 1995)	×	×	bootstrap only	✓	✓
A-SIMEX (this paper)	✓ ^e	✓	✓	✓	✓

Notes. SIMEX extensions to misclassification and semiparametric settings: Küchenhoff et al. (2006) and Apanasovich et al. (2009), respectively. Deconvolution rates: Fan and Truong (1993). ^aconsistent for the naive pseudo-parameter $\beta(1)$, not for β_0 (Lemma 1). ^bapproximate bias correction whose error is first order in the error variance (Section 5.1). ^cRMSE deteriorates sharply when many coefficients are weakly identified (Section 5.3). ^dposterior consistency known in special cases. ^ethe CV criterion systematically under-extrapolates when the error is large, a limitation we document and quantify rather than suppress (Section 5.1).

deviance, with a one-standard-error parsimony rule. This removes subjectivity and is fully reproducible. Our simulations, however, reveal a structural limitation that we do not paper over: the population minimizer of the observed-scale CV criterion is the naive pseudo-parameter $\beta(1)$, not β_0 , so CV systematically under-extrapolates when the measurement error is large (Section 5.1). Automatic tuning is a solution for small-to-moderate error and a documented failure mode for large error, for which we propose deconvolution-aware remedies.

2. **A bias-corrected two-stage extrapolation and an exact variance estimator.** The primary polynomial fit is stabilized by a ridge second stage, and inference uses an analytic sandwich that decomposes the asymptotic variance exactly into a *data* component and a *simulation* component of order B^{-1} ; this decomposition corrects the additive “parameter-plus-extrapolation” split that appears in earlier SIMEX treatments, and it extends robustly to overdispersed outcomes and to estimated error variance (Sections 5.4 and 5.5).
3. **Complete and corrected asymptotic theory.** We prove consistency and asymptotic normality of the A-SIMEX estimator, a growing-degree central limit theorem that reconciles the $\log n$ degree growth needed for consistency with centered inference through an explicit window condition on the analyticity margin, and oracle-efficiency results comprising an information-monotonicity lemma (no estimator based on noisy covariates beats the oracle to first order) and an explicit second-order expansion whose sign is design-dependent. Along the way we correct three claims that circulate informally for SIMEX: the exactness of the extrapolation target, the form of the variance decomposition, and the sign of the second-order efficiency loss.
4. **Evidence at replication standard.** All simulations (1,000 Monte Carlo repetitions per

cell) and the application are generated by a single seed-fixed pipeline; the paper reports what the pipeline produces, including results that are unflattering to the method.

The methods are implemented in an open-source R package, and every number in the paper is regenerable from a documented archive with fixed seeds. The plan is as follows. Section 2 reviews the model and standard SIMEX. Section 3 presents A-SIMEX: cross-validated selection of $(\lambda_{\max}, \text{degree})$ (Section 3.1), the ridge second stage (Section 3.2), and the analytic sandwich variance estimator (Section 3.3). Section 4 states the main results: the population SIMEX curve and its analyticity (Lemma 1), the consistency theorem (Theorem 1), asymptotic normality with the exact variance decomposition (Theorem 2, Corollary 1), the growing-degree central limit theorem (Theorem 3), and the oracle-efficiency results (Lemma 6, Theorem 4). Section 5 reports the simulation studies; Section 6 the application to NHANES 2017–2018; Section 8 the software; Section 9 limitations and extensions. Complete proofs are in the Supplementary Material.

2. Background and Setup

2.1. Poisson Regression

We observe n independent subjects $i = 1, \dots, n$ with count outcome $Y_i \in \mathbb{N}_0$ and covariate vector $X_i \in \mathbb{R}^p$. The Poisson regression model specifies

$$Y_i \mid X_i \sim \text{Poisson}(\mu_i), \quad \mu_i = \exp(X_i^\top \beta_0), \quad (1)$$

where $\beta_0 \in \Theta \subset \mathbb{R}^p$ is the parameter of interest, with log-likelihood

$$\ell_n(\beta) = \sum_{i=1}^n \left\{ Y_i X_i^\top \beta - \exp(X_i^\top \beta) \right\}. \quad (2)$$

The oracle maximum likelihood estimator $\hat{\beta}_{\text{oracle}}$, infeasible because it requires X_i , solves $\nabla \ell_n(\beta) = \mathbf{0}$ and is asymptotically normal with variance equal to the inverse Fisher information.

2.2. Classical Measurement Error Model

In practice we observe only a noisy version,

$$W_i = X_i + \varepsilon_i, \quad (3)$$

with $\varepsilon_i \sim N(\mathbf{0}, \Sigma_\varepsilon)$ independent of (X_i, Y_i) and $\Sigma_\varepsilon = \text{diag}(\sigma_{\varepsilon,1}^2, \dots, \sigma_{\varepsilon,p}^2)$ assumed known or consistently estimable from auxiliary data (validation subsamples or replicated measurements; Section 5.5 implements the latter).

Assumption 1 (Classical error structure). $\varepsilon_i \sim N(\mathbf{0}, \Sigma_\varepsilon)$ independently across i , independent of (X_i, Y_i) ; Σ_ε is known (or consistently estimated).

2.3. The Naive Estimator and Its Bias

Fitting the Poisson GLM with W_i in place of X_i yields the naive MLE $\hat{\beta}_{\text{naive}}$, which is attenuated toward zero. The attenuation is characterized exactly in Lemma 1(iii) below: the pseudo-true parameter of the naive fit is $\beta(1) = \beta_0 + \tilde{\beta} + O(\|\Sigma_\epsilon\|^2)$, with first-order bias

$$\tilde{\beta} = -\mathbf{I}_X(\beta_0)^{-1} \left[\frac{1}{2} (\beta_0^\top \Sigma_\epsilon \beta_0) \mathbb{E}(e^{X_i^\top \beta_0} X_i) + \Sigma_\epsilon \beta_0 \mathbb{E}(e^{X_i^\top \beta_0}) \right], \quad (4)$$

where $\mathbf{I}_X(\beta_0) = \mathbb{E}[e^{X_i^\top \beta_0} X_i X_i^\top]$ is the oracle information. For $p = 1$, $X \sim N(0, 1)$ this gives $\tilde{\beta} = -\sigma_\epsilon^2 \beta_0 (1 + \beta_0^2/2)/(1 + \beta_0^2) < 0$ for $\beta_0 > 0$: attenuation, with the same first-order structure the linear model is famous for but governed by the moment-generating function of X at β_0 rather than by a closed-form dilution factor.

2.4. Standard SIMEX

Simulation extrapolation (Cook and Stefanski, 1994) augments the observed error artificially and extrapolates backward:

Algorithm 1 (Standard SIMEX)

Require: data (Y_i, W_i) ; assumed Σ_ϵ ; grid $\Lambda = \{\lambda_0 = 0 < \dots < \lambda_K\}$; B replicates.
1 **for** each $\lambda \in \Lambda$, $b = 1, \dots, B$:
2 draw $Z_{ib} \sim N(\mathbf{0}, \Sigma_\epsilon)$ i.i.d., set $W_i^{(b)}(\lambda) = W_i + \sqrt{\lambda} Z_{ib}$;
3 fit the Poisson GLM of Y_i on $W_i^{(b)}(\lambda)$, obtaining $\hat{\beta}_b(\lambda)$;
4 average: $\bar{\beta}(\lambda) = B^{-1} \sum_b \hat{\beta}_b(\lambda)$.
5 **Extrapolate:** fit $p(\lambda) = a + b\lambda + c\lambda^2$ to $(\lambda_j, \bar{\beta}(\lambda_j))$ and report $\hat{\beta}_{\text{SIMEX}} = p(-1)$.

The intuition: increasing λ compounds the error, moving $\bar{\beta}(\lambda)$ further from β_0 along a smooth bias curve; extrapolating to $\lambda = -1$ “removes” the original error. Section 4.2 makes this precise: the population curve $\beta(\lambda)$ depends on the error only through the *total* variance $(1 + \lambda)\Sigma_\epsilon$, is real-analytic in λ , and satisfies $\beta(-1) = \beta_0$ exactly, but the naive reading “ $W(-1) = W - \epsilon = X$ ” is wrong (the simulated noise is independent of the observed error and cannot be subtracted), a distinction that matters for every formal claim.

2.5. Limitations of Standard SIMEX

1. *Ad-hoc hyperparameters:* the grid Λ and the polynomial degree are chosen by hand, with no principled guidance and consequences for reproducibility.
2. *Extrapolation instability:* polynomial extrapolation from $[0, \lambda_{\max}]$ to $\lambda = -1$ amplifies noise (by a factor governed by the weights $\sum_j |c_j|$; see Theorem 4), and high-degree polynomials oscillate outside the fitted range.
3. *No formal asymptotics:* the bias, variance, and coverage properties of $\hat{\beta}_{\text{SIMEX}}$ had not been rigorously characterized for GLMs; existing analyses are heuristic.
4. *Computational cost:* $B \times |\Lambda|$ GLM fits per evaluation, burdensome when p is moderate, though The Supplementary Material (Section S2) shows that batching and warm starts reduce this to seconds.

2.6. Related Work

Corrected score equations (Nakamura, 1990) replace the score by an estimate whose conditional expectation given (X, Y) is the true score; they require the error distribution (as SIMEX does) rather than a model for $\Pr(X | W)$, but their variance explodes when many coefficients are weakly identified (Section 5.3). Deconvolution and functional methods (Fan and Truong, 1993; Delaigle and Meister, 2008) estimate the density of X from W nonparametrically; they are elegant but ill-posed and hard to scale. Empirical likelihood (Qin, 2000) can incorporate validation data but requires the validation sample itself. Our empirical comparisons include the first two families (Sections 5.1–5.3); the variance estimation from replicated measures covers the third’s typical data structure.

3. Adaptive SIMEX: Method and Implementation

3.1. Data-Driven Hyperparameter Selection via Cross-Validation

The core innovation of A-SIMEX is to treat the extrapolation region and polynomial degree as tuning parameters selected by k -fold cross-validation using prediction loss on held-out data.

Cross-validation objective

Partition the data into K_{fold} folds (typically five). For a candidate pair $(\lambda_{\text{max}}, p_{\text{deg}})$:

Fit stage. For fold k , using data excluding fold k , run Algorithm 1 with extrapolation grid $\{0, 0.5, 1, \dots, \lambda_{\text{max}}\}$ and degree p_{deg} , obtaining the extrapolated estimate $\hat{\beta}^{(-k)}$. Within a fold, the curve on the union grid is fitted once and each candidate λ_{max} is evaluated on the corresponding prefix (the *prefix property*), with noise draws shared across λ within a replicate.

Prediction stage. On the held-out fold, compute the Poisson deviance of the fitted counts $\hat{Y}_{i,k} = \exp(W_{i,k}^\top \hat{\beta}^{(-k)})$:

$$D_k(\lambda_{\text{max}}, p_{\text{deg}}) = 2 \sum_{i \in \text{fold } k} \left[Y_i \log \frac{Y_i}{\hat{Y}_{i,k}} - (Y_i - \hat{Y}_{i,k}) \right], \quad (5)$$

with the convention $0 \log 0 = 0$, and aggregate $\text{CV}(\lambda_{\text{max}}, p_{\text{deg}}) = K_{\text{fold}}^{-1} \sum_k D_k$. The selected pair minimizes CV; when candidates lie within one standard error of the minimum, the *parsimonious* choice (smaller degree, then smaller λ_{max}) is preferred. Section 5.1 quantifies a bias of this criterion toward under-extrapolation and Section 5.6 discusses remedies; the selected pair is additionally reported in all output so that practitioners can inspect the choice.

Computational efficiency

The CV procedure is parallelizable at two levels (across folds; across grid points), and the prefix property reduces the number of fitted pseudo-samples per fold from $|\mathcal{K}_\lambda| \cdot |\Lambda| B_{\text{cv}}$ to $|\Lambda| B_{\text{cv}}$ (Supplementary Material, Section S2).

3.2. Bias-Corrected Extrapolation

Even with optimal $(\lambda_{\text{max}}, p_{\text{deg}})$, the polynomial extrapolation to $\lambda = -1$ may suffer from approximation bias. We use a two-stage procedure. Stage 1 fits the primary degree- d polynomial $p(\lambda) = \phi(\lambda)^\top \theta$ with $\phi(\lambda) = (1, \lambda, \dots, \lambda^d)^\top$ to the $(\lambda_j, \bar{\beta}(\lambda_j))$ points. Stage 2 estimates a residual correction by ridge regression of the primary-fit residuals on a richer basis $\phi_{d+2}(\lambda)$:

$$\hat{r}_{\text{ridge}}(\lambda) = \phi_{d+2}(\lambda)^\top \arg \min_{\theta_r} \left\| \bar{\beta} - \mathbf{P} \bar{\beta} - \Phi_r \theta_r \right\|^2 + \gamma \|\theta_r\|^2, \quad (6)$$

where $\mathbf{P}$ is the projection onto the span of the primary basis, Φ_r the $(K \times (d+3))$ basis matrix, and $\gamma = 0.01$; the final estimate is $\hat{\beta}_{\text{A-SIMEX}} = p(-1) + \hat{r}_{\text{ridge}}(-1)$. Because both stages are linear in the curve, the estimate is a single linear functional $\hat{\beta}_{\text{A-SIMEX}} = \sum_j d_j \bar{\beta}(\lambda_j)$ with combined weights $d = c + c_{\text{ridge}}$ satisfying $\sum_j d_j = 1$, the form in which the variance estimator below and the asymptotic theory analyze it. The ridge stage is $O_p(B^{-1/2})$ at the $\sqrt{n}$ scale (the theory treats it exactly through the combined weights).

3.3. Variance Estimation and Confidence Intervals

Let $c = (c_0, \dots, c_{K-1})$ denote the combined extrapolation weights (primary polynomial plus ridge second stage), so that the A-SIMEX estimator is exactly the linear functional $\hat{\beta}_{\text{A-SIMEX}} = \sum_j c_j \bar{\beta}(\lambda_j)$ with $\sum_j c_j = 1$, and let $\mathbf{H}_j = \mathbb{E}[e^{\beta(\lambda_j)^\top W(\lambda_j)} W(\lambda_j) W(\lambda_j)^\top]$ be the per-observation Hessian of the population SIMEX objective at grid point λ_j . Under the conditions of Theorem 2,

$$\text{Var}(\hat{\beta}_{\text{A-SIMEX}}) \approx \frac{1}{n} \mathbf{C} (\mathbf{V}^{\text{data}} + \frac{1}{B} \mathbf{V}^{\text{sim}}) \mathbf{C}^\top, \quad \mathbf{C} = [c_0 \mathbf{H}_0^{-1}, \dots, c_{K-1} \mathbf{H}_{K-1}^{-1}], \quad (7)$$

where $\mathbf{V}^{\text{data}}$ is the covariance across observations of the stacked conditional influence functions $\mathbb{E}[\mathbf{m}_{ib}(\lambda_j) \mid Y_i, \mathbf{w}_i]$ and $\mathbf{V}^{\text{sim}}$ their conditional covariance given the data; both admit closed-form expressions under the Gaussian error model (Gaussian tilt identities) and are estimated from the same replicates that produce the SIMEX curve, at negligible additional cost. Two features distinguish (7) from the additive “parameter-plus-extrapolation” split that appears in earlier SIMEX treatments: (i) the naive Fisher information $\mathbf{I}_W(\beta_0)^{-1}$ does *not* enter additively: the estimator is a fixed weighted average of the grid-point pseudo-estimators, and its variance is the propagated influence covariance (7), which reduces to the naive variance only in the limit of perfectly correlated grid-point influence; and (ii) the simulation noise enters at exactly $O(B^{-1})$, so $B = 100$ makes it negligible relative to the data component, as confirmed empirically in Section 5.6, where the analytic sandwich recovers 66–91% of the parametric-bootstrap variance (first-order under-statement, improving with n). Confidence intervals are $\hat{\beta}_{\text{A-SIMEX},j} \pm z_{1-\alpha/2} \widehat{\text{se}}_j$ from (7); under overdispersion the same sandwich is valid with robust scores, and Section 5.4 verifies this under negative-binomial outcomes.

4. Asymptotic Theory

This section states the theoretical properties of the A-SIMEX estimator; complete proofs are collected in the Supplementary Material. Throughout, $\|\cdot\|$ is the Euclidean norm, $\rightarrow_p$ denotes convergence in probability, $\rightarrow_d$ convergence in distribution, and generic positive constants are written C, c .

4.1. Setup and Assumptions

Assumption 2 (Model and regularity).

- (a) $Y_1, X_1, \varepsilon_1, \dots, Y_n, X_n, \varepsilon_n$ are i.i.d.; $\mathbb{E}[Y_i \mid X_i] = \exp(X_i^\top \beta_0)$ with β_0 in the interior of the compact convex set $\Theta \subset \mathbb{R}^p$.
- (b) $\mathbb{E}\|X_i\|^4 < \infty$, $\mathbb{E}|Y_i|^2 < \infty$, and $\mathbb{E}[X_i X_i^\top]$ is positive definite.
- (c) The error satisfies Assumption 1.

Assumption 3 (Pseudo-design non-collinearity). For every δ in a neighbourhood of $[-1, \lambda_{\max} + 1]$ and every β in a neighbourhood of β_0 ,

$$\mathbf{H}(\beta, \delta) := \mathbb{E} \left[\exp(\beta^\top (X_i + \varepsilon_\delta)) (X_i + \varepsilon_\delta) (X_i + \varepsilon_\delta)^\top \right]$$

is positive definite, where $\varepsilon_\delta \sim N(\mathbf{0}, \delta \Sigma_\varepsilon)$; in particular $\inf_{\beta, \delta} \lambda_{\min}(\mathbf{H}(\beta, \delta)) \geq c_H > 0$.

Assumption 4 (Simulation and grid). The number of simulation replicates satisfies $B = B(n) \rightarrow \infty$. The grid Λ_n has $K = K(n)$ equally spaced points on $[0, \lambda_{\max}]$ with $\lambda_{\max} \leq \bar{\lambda} < \infty$ fixed, $K \rightarrow \infty$, and $K^2 \log n = o(n)$.

Assumption 5 (Extrapolation degree; consistency regime). The degree grid is $\{1, \dots, d_{\max}(n)\}$ with $d_{\max}(n) \rightarrow \infty$, $d_{\max}(n) \leq K(n)/4$, and the selected degree $\hat{d}_n$ satisfies $\hat{d}_n \rightarrow \infty$ in probability and $d_{\max}(n)^3/K(n) \rightarrow 0$.

Assumption 6 (Selection stability; central limit regime). The CV procedure selects $(\hat{\lambda}_{\max, n}, \hat{d}_n)$ from a fixed finite grid $\mathcal{G}$ with $\mathbb{P}((\hat{\lambda}_{\max, n}, \hat{d}_n) = (\lambda^*, d^*)) \rightarrow 1$ for some $(\lambda^*, d^*) \in \mathcal{G}$.

Remark 1. Assumptions 5 (degree grows: consistency) and 6 (degree eventually constant: the CLT) describe complementary regimes; Theorem 3 reconciles them under an explicit window condition on a $c \log n$ degree growth rate.

Remark 2. Three of these conditions correct or replace claims made in informal treatments. First, no condition of the form $B/\sqrt{n} \rightarrow \infty$ (with $B/n \rightarrow 0$) is needed: the simulation component of the estimator is $O_p(B^{-1/2})$ at the $\sqrt{n}$ scale, so $B \rightarrow \infty$ suffices for all results. Second, a fixed polynomial degree cannot deliver consistency in general: the extrapolation approximation error does not vanish with $\lambda_{\max}$, which is why Assumption 5 lets the CV-selected degree grow; real-analyticity of the population curve (Lemma 1) then makes the approximation error decay geometrically in the degree. Third, selection stability (Assumption 6) replaces “convergence with positive probability” of the selected hyperparameters, which is too weak for a central limit theorem; the cross-validation criterion supports both regimes under verifiable conditions on the population risk (Lemma 4).

4.2. The Population SIMEX Curve

Write $\delta = 1 + \lambda$ for the total error-variance multiplier and $\varepsilon_\delta = \varepsilon_i + \sqrt{\lambda} \mathbf{u}_i \sim N(\mathbf{0}, \delta \Sigma_\varepsilon)$ for the augmented error. Define the population objective and pseudo-true parameter

$$Q(\beta; \delta) = \mathbb{E} \left[Y_i \beta^\top (X_i + \varepsilon_\delta) - \exp(\beta^\top (X_i + \varepsilon_\delta)) \right], \quad \beta(\delta) = \arg \max_{\beta \in \Theta} Q(\beta; \delta), \quad (8)$$

and write $\beta(\lambda) := \beta(1 + \lambda)$ for the population SIMEX curve.

Lemma 1 (Population representation, analyticity, and the extrapolation target). *Grant Assumptions 2 and 3. Then:*

(i) Q depends on δ only through

$$Q(\beta; \delta) = \mathbb{E}[Y_i X_i^\top \beta] - \mathbb{E}[\exp(X_i^\top \beta)] \exp\left(\frac{\delta}{2} \beta^\top \Sigma_\varepsilon \beta\right),$$

hence $Q(\cdot; \delta)$ is real-analytic in (β, δ) on $\text{int } \Theta \times (-1 - \eta, \bar{\lambda} + 1)$ for some $\eta > 0$.

- (ii) The maximizer $\beta(\delta)$ exists, is unique, lies in the interior of Θ , and is real-analytic in δ on a neighbourhood of $[-1, \bar{\lambda} + 1]$.
- (iii) $\beta(-1) = \beta(0) = \beta_0$; moreover $\beta(\delta) = \beta_0 + \delta \tilde{\beta} + O(\delta^2)$ as $\delta \downarrow 0$, where, writing $\mathbf{g}(\beta) = \mathbb{E}[e^{X_i^\top \beta} X_i]$ and $M(\beta) = \mathbb{E}[e^{X_i^\top \beta}]$,

$$\tilde{\beta} = -\mathbf{H}(\beta_0, 0)^{-1} \left[\frac{1}{2} (\beta_0^\top \Sigma_\epsilon \beta_0) \mathbf{g}(\beta_0) + \Sigma_\epsilon \beta_0 M(\beta_0) \right], \quad (9)$$

the leading-order bias of the naive estimator. For $p = 1$, $X \sim N(0, 1)$: $\tilde{\beta} = -\sigma_\epsilon^2 \beta_0 (1 + \beta_0^2/2)/(1 + \beta_0^2) < 0$ for $\beta_0 > 0$.

Two remarks on Lemma 1. First, part (iii) is the correct form of the intuition that “at $\lambda = -1$ the augmented covariate equals X ”: the simulated noise is independent of the observed error and can never be subtracted off; instead, the augmented covariate at level λ is $X + \epsilon_\delta$ with *total* error variance $\delta \Sigma_\epsilon$, and the pseudo-true parameter depends on δ only through this total variance, so $\lambda = -1$ corresponds to a pseudo-population with no error at all, giving $\beta(-1) = \beta_0$ exactly. Second, part (i) is what drives every approximation result below: the Gaussian tilt makes the population curve real-analytic, so polynomial approximation error decays geometrically in the degree (Lemma 3).

4.3. Uniform Convergence of the Empirical Curve

Let $Q_n(\beta; \lambda, b) = n^{-1} \sum_{i=1}^n [Y_i W_i^{(b)}(\lambda)^\top \beta - \exp\{W_i^{(b)}(\lambda)^\top \beta\}]$ with $W_i^{(b)}(\lambda) = W_i + \sqrt{\lambda} U_{ib}$ and $\{U_{ib}\}$ i.i.d. $N(\mathbf{0}, \Sigma_\epsilon)$ across i and b ; $\hat{\beta}_b(\lambda) = \arg \max_{\beta \in \Theta} Q_n(\beta; \lambda, b)$; $\bar{\beta}(\lambda) = B^{-1} \sum_b \hat{\beta}_b(\lambda)$; and $\mathbf{H}(\lambda) := \mathbf{H}(\bar{\beta}(\lambda), 1 + \lambda)$.

Lemma 2 (Uniform M-estimation over grid and replicates). *Grant Assumptions 2–4. Then*

$$\max_{0 \leq j < K} \max_{1 \leq b \leq B} \sup_{\beta \in \Theta} |Q_n(\beta; \lambda_j, b) - Q(\beta; 1 + \lambda_j)| \xrightarrow{p} 0,$$

and consequently $\max_j \|\bar{\beta}(\lambda_j) - \beta(\lambda_j)\| = O_p(n^{-1/2})$ up to polylogarithms, with each $\hat{\beta}_b(\lambda_j)$ well defined and unique with probability tending to one.

Lemma 3 (Extrapolation approximation error). *Grant Assumptions 2–4. There exist constants $C, \rho > 1$ depending only on the law of (X_i, Y_i) , Σ_ϵ , $\bar{\lambda}$ and the grid such that for all $d \geq 1$,*

$$\inf_{\deg(q) \leq d} \sup_{\lambda \in [-1, \bar{\lambda}]} \|q(\lambda) - \beta(\lambda)\| \leq C \rho^{-d}.$$

Consequently, under Assumption 5, the best degree- $\hat{d}_n$ approximation error is $o_p(n^{-1/2})$ whenever $d_{\max}(n) \geq c \log n$ for large enough c .

Lemma 4 (Cross-validation and degree growth). *Let $\hat{R}_n(\lambda_{\max}, d)$ be the CV criterion (5) over the finite grid $\mathcal{G}_n = \Lambda_n^{\max} \times \{1, \dots, d_{\max}(n)\}$ and R_n the population CV risk. If (i) $\sup_{g \in \mathcal{G}_n} |\hat{R}_n(g) - R_n(g)| = o_p(1)$; (ii) $R_n(\lambda_{\max}, d) \geq \rho^{-2d}(1 - o(1))$ uniformly over $d \leq d_0$ for every fixed d_0 ; and (iii) R_n is minimized over d at some $\bar{d}_n \rightarrow \infty$, then the CV-selected degree $\hat{d}_n \rightarrow \infty$ in probability. If moreover the grid contains a fixed pair with a strict population minimum separated by more than $2 \sup_g |\hat{R}_n(g) - R_n(g)| = o_p(1)$, then Assumption 6 holds.*

4.4. Consistency

Theorem 1 (Consistency of A-SIMEX). *Grant Assumptions 2–5. Then $\hat{\beta}_{\text{A-SIMEX}} \xrightarrow{p} \beta_0$. If $d_{\max}(n) \geq c \log n$ eventually, then $\|\hat{\beta}_{\text{A-SIMEX}} - \beta_0\| = O_p(\rho^{-\hat{d}_n}) + O_p(\sqrt{d_{\max}(n)/K(n)} n^{-1/2}) + O_p(n^{-1/2})$.*

4.5. Asymptotic Normality and the Exact Variance Decomposition

For the central limit theorem we work in the fixed-hyperparameter regime of Assumption 6. To lighten notation write $d = d^*$, $\lambda_{\max} = \lambda^*$, and $\phi(\lambda) = (1, \lambda, \dots, \lambda^d)^\top$. Define the per-replicate influence score at grid point λ_j ,

$$\mathbf{m}_{ib}(\lambda_j) = \left[Y_i - \exp(\beta(\lambda_j)^\top W_i^{(b)}(\lambda_j)) \right] W_i^{(b)}(\lambda_j),$$

whose conditional mean given $(Y_i, X_i, \varepsilon_i)$ does not depend on b and has zero unconditional mean at $\beta(\lambda_j)$; and the matrices

$$\mathbf{V}^{\text{data}} = \text{Cov} \left(\mathbb{E}[\mathbf{m}_{ib}(\lambda_j) \mid Y_i, X_i, \varepsilon_i] \right)_{j,k=0}^{K-1} \in \mathbb{R}^{Kp \times Kp}, \quad (10)$$

$$\mathbf{V}^{\text{sim}} = \text{Cov} \left(\mathbf{m}_{ib}(\lambda_j) - \mathbb{E}[\mathbf{m}_{ib}(\lambda_j) \mid Y_i, X_i, \varepsilon_i] \right)_{j,k} \in \mathbb{R}^{Kp \times Kp}, \quad (11)$$

the covariance contributed by the data and by the simulated errors, stacked across the grid. Let $\mathbf{H} = \text{blockdiag}(\mathbf{H}(\lambda_0), \dots, \mathbf{H}(\lambda_{K-1}))$, $\mathbf{M} = \mathbf{I}_p \otimes [(\Phi^\top \Phi)^{-1} \Phi^\top]$ the least-squares extrapolation map, and $\mathbf{v}_{-1} := \mathbf{I}_p \otimes \phi(-1)$, so that $\hat{\beta}_{\text{A-SIMEX}} = \mathbf{v}_{-1}^\top \mathbf{M} \bar{\beta}_\Lambda$ with $\bar{\beta}_\Lambda = (\bar{\beta}(\lambda_0)^\top, \dots, \bar{\beta}(\lambda_{K-1})^\top)^\top$.

Theorem 2 (Asymptotic normality of A-SIMEX). *Grant Assumptions 2–3 and 6, with $B \rightarrow \infty$ and fixed $K, \lambda_{\max}, d$. Then*

$$\sqrt{n}(\hat{\beta}_{\text{A-SIMEX}} - \beta_0) \xrightarrow{d} N(\mathbf{0}, \Sigma_{\text{A-SIMEX}}),$$

with the exact decomposition

$$\Sigma_{\text{A-SIMEX}} = \Sigma_{\text{data}} + \frac{1}{B} \Sigma_{\text{sim}}, \quad \Sigma_{\text{data}} = \mathbf{v}_{-1}^\top \mathbf{M} \mathbf{H}^{-1} \mathbf{V}^{\text{data}} \mathbf{H}^{-1} \mathbf{M}^\top \mathbf{v}_{-1}, \quad (12)$$

and Σ_{sim} defined analogously with $\mathbf{V}^{\text{sim}}$ in place of $\mathbf{V}^{\text{data}}$. In particular the simulation component vanishes for any rate $B \rightarrow \infty$, including $B = o(n)$.

Remark 3 (Correction to the additive decomposition). The asymptotic variance is the propagated influence covariance (12) of the pseudo-estimators at the grid points. An additive split $\Sigma_{\text{param}} + \Sigma_{\text{extrap}}$ with $\Sigma_{\text{param}} = \mathbf{I}_W(\beta_0)^{-1}$ (the information of the naive working model) does not hold: the A-SIMEX estimator is a fixed weighted average $\sum_j c_j \bar{\beta}(\lambda_j)$ of the grid-point estimators with $\sum_j c_j = 1$, the grid-point influence functions are strongly (asymptotically perfectly, for small error) correlated, and the naive information enters only through this correlation structure. The correct additive split is between data and simulation sources of randomness, with the simulation term exactly $O(B^{-1})$.

Corollary 1 (Variance estimation). *Under the conditions of Theorem 2, the plug-in estimator described in Section 3.3 (sample-analogue $\mathbf{H}_j$ from the replicate-level information matrices, and empirical covariance matrices of the estimated conditional influence means and residuals) is consistent for $\Sigma_{\text{A-SIMEX}}$; equivalently, the pairs-bootstrap (redrawing both data and simulated errors) is consistent.*

4.6. The Growing-Degree Regime

Map $[0, \bar{\lambda}]$ affinely to $[-1, 1]$ by $t(\lambda) = 2\lambda/\bar{\lambda} - 1$, so that $\lambda = -1$ maps to $t_0 = -1 - 2/\bar{\lambda} < -1$, and define the extrapolation conditioning constant

$$\kappa := |t_0| + \sqrt{t_0^2 - 1} = 1 + \frac{2}{\bar{\lambda}} + \sqrt{\left(1 + \frac{2}{\bar{\lambda}}\right)^2 - 1} > 1,$$

the Bernstein parameter of t_0 : by the Chebyshev extremal property, any polynomial q of degree k satisfies $|q(t_0)| \leq \kappa^k \sup_{[-1, 1]} |q|$ (Rivlin, 1969).

Lemma 5 (Extrapolation operator bounds). *Let $K \geq c_0 d^2$ equispaced points on $[0, \bar{\lambda}]$, $\{\varphi_k\}_{k=0}^d$ the polynomials orthonormal with respect to the discrete inner product $K^{-1} \sum_j f(\lambda_j)g(\lambda_j)$, and $L_{K,d}(h) = \sum_j c_j h(\lambda_j)$ the “least-squares degree- d fit evaluated at $\lambda = -1$ ” functional. Then: (i) $\sum_{k=0}^d \varphi_k(t_0)^2 \leq C d \kappa^{2d}$; (ii) $\sum_j |c_j| \leq C d^{1/2} K \kappa^d$ and $L_{K,d}(q) = q(-1)$ for every polynomial of degree $\leq d$; (iii) for every h real-analytic on a neighbourhood of $[-1, \bar{\lambda}]$ with analytic continuation bounded by M on the Bernstein ellipse E_ρ , $|L_{K,d}(h) - h(-1)| \leq C M d^{3/2} K \kappa^d \rho^{-d}$.*

Theorem 3 (Growing-degree central limit theorem). *Grant Assumptions 2–4 with $K(n) \geq c_0 d_n^2$ equispaced points and $B \rightarrow \infty$. Suppose the selected degree satisfies $\hat{d}_n/d_n \xrightarrow{P} 1$ for a deterministic sequence $d_n = \lceil c \log n \rceil$, and that the analyticity margin $\rho > \kappa$ holds with $c \in (\frac{1}{2 \log \rho}, \frac{1}{2 \log \kappa})$, an interval that is nonempty exactly when $\rho > \kappa$. Then*

$$\sqrt{n}(\hat{\beta}_{\text{A-SIMEX}} - \beta_0) \xrightarrow{d} N(\mathbf{0}, \Sigma_*), \quad \Sigma_* = \mathbf{I}_X(\beta_0)^{-1} \tilde{\mathbf{Z}} \mathbf{I}_X(\beta_0)^{-1},$$

where $\tilde{\mathbf{Z}}$ is the value at $(-1, -1)$ of the analytic continuation of the influence covariance $\mathbf{\Gamma}(\lambda, \lambda') = \text{Cov}(\psi_1(\lambda), \psi_1(\lambda'))$ of the data-influence process $\psi(\lambda) = \mathbb{E}[\mathbf{m}_{ib}(\lambda) \mid Y_i, X_i, \boldsymbol{\varepsilon}_i]$, and $\mathbf{I}_X(\beta_0) = \mathbf{H}(-1)$. The simulation component contributes $O(B^{-1})$, and the extrapolation bias is $o(n^{-1/2})$, so the limit is centered at β_0 . In particular $\Sigma_* \rightarrow \Sigma_{\text{oracle}}$ as $\sigma^2 \rightarrow 0$.

The window condition has a transparent reading: ρ measures how far the analytic continuation of the curve extends (bias side), κ how violently degree- d extrapolation outside the grid amplifies noise (variance side); growing-degree inference is feasible precisely when the former dominates the latter. If $\rho \leq \kappa$, the fixed-degree central limit theorem (Theorem 2) is the appropriate inference tool.

4.7. Efficiency Relative to the Oracle

Let $\Sigma_{\text{oracle}} = \mathbf{I}_X(\beta_0)^{-1}$ and define the generalized asymptotic relative efficiency $\text{ARE} = \det(\Sigma_{\text{oracle}})/\det(\Sigma_{\text{A-SIMEX}})$. Three information matrices must be distinguished: the oracle information $\mathbf{I}_X(\beta_0)$; the information $\mathbf{I}(t)$ of the observed-data model $(Y_i, \tilde{W}_t)$, $\tilde{W}_t = X_i + \mathbf{e}_t$ with $\mathbf{e}_t \sim N(\mathbf{0}, t\Sigma_0)$; and the information of the naive *working* model $\mathbb{E}[e^{W^\top \beta_0} W W^\top]$, a curvature of a misspecified likelihood that plays no role in the correct theory.

Lemma 6 (Information monotonicity under additive noise). *Grant Assumption 2 with Σ_0 fixed. Then: (i) $\mathbf{I}(t_2) \preceq \mathbf{I}(t_1)$ for $0 \leq t_1 \leq t_2$; in particular $\mathbf{I}(t) \preceq \mathbf{I}(0) = \mathbf{I}_X(\beta_0)$; (ii) the inequality is strict for $t > 0$ provided X_i is not a.s. constant. Consequently $\Sigma_{\text{oracle}} \preceq \mathbf{I}(t)^{-1}$ for all $t \geq 0$, and by the Hájek convolution theorem $\mathbf{I}(t)^{-1}$ is the best asymptotic covariance achievable by any regular estimator based on $(Y_i, \tilde{W}_t)$ alone.*

Theorem 4 (Efficiency relative to the oracle). *Grant the conditions of Theorem 2 with $B \rightarrow \infty$, and suppose $\Sigma_\epsilon = \sigma^2 \Sigma_0$ with Σ_0 fixed and $\sigma^2 \downarrow 0$, all else fixed. Write $\Sigma_{\text{A-SIMEX}}(\sigma^2)$ for the asymptotic covariance of Theorem 2 at error scale σ^2 . Then:*

- (i) **(First-order efficiency.)** $\Sigma_{\text{A-SIMEX}}(\sigma^2) \rightarrow \Sigma_{\text{oracle}}$ as $\sigma^2 \rightarrow 0$, for every fixed grid and degree: the limit is exact.
- (ii) **(Explicit second-order term.)** $\Sigma_{\text{A-SIMEX}}(\sigma^2) = \Sigma_{\text{oracle}} + \sigma^2 \dot{C}_\beta + O(\sigma^4)$, where $\dot{C}_\beta$ is the symmetric, explicitly computable matrix

$$\dot{C}_\beta = \mathbf{v}_{-1}^\top \mathbf{M} \mathbf{H}_0^{-1} (\dot{\mathbf{V}} - \dot{\mathbf{H}} \mathbf{H}_0^{-1} \mathbf{V}_0 - \mathbf{V}_0 \mathbf{H}_0^{-1} \dot{\mathbf{H}}) \mathbf{H}_0^{-1} \mathbf{M}^\top \mathbf{v}_{-1}, \quad (13)$$

with $\mathbf{H}_0 = \mathbf{I}_K \otimes \mathbf{I}_X(\beta_0)$, $\mathbf{V}_0 = (\mathbf{1}_K \mathbf{1}_K^\top) \otimes \mathbf{I}_X(\beta_0)$, and

$$\dot{\mathbf{H}}_j = \frac{\delta_j}{2} (\beta_0^\top \Sigma_0 \beta_0) \mathbf{I}_X + \delta_j \mathbb{E}[e^{\beta_0^\top X} (\tilde{\beta}_{\Sigma_0}^\top X) X X^\top] + \Sigma_0 \beta_0 \mathbf{g}_0^\top + \mathbf{g}_0 \beta_0^\top \Sigma_0, \quad (14)$$

$$\dot{\mathbf{V}}_{jk} = \text{Cov}(\dot{\psi}_j, \psi^{(0)}) + \text{Cov}(\psi^{(0)}, \dot{\psi}_k),$$

$$\dot{\psi}_j = -e^{\beta_0^\top X} \left[(\delta_j \tilde{\beta}_{\Sigma_0}^\top X + \frac{\lambda_j}{2} \beta_0^\top \Sigma_0 \beta_0) X + \lambda_j \Sigma_0 \beta_0 \right], \quad (15)$$

where $\delta_j = 1 + \lambda_j$, $\mathbf{g}_0 = \mathbb{E}[e^{\beta_0^\top X} X]$, and $\psi^{(0)} = YX - e^{\beta_0^\top X} X$ is the oracle influence function. In general $\dot{C}_\beta$ is indefinite: neither $\dot{C}_\beta \succeq \mathbf{0}$ nor $\dot{C}_\beta \preceq \mathbf{0}$ holds universally.

- (iii) **(Determinant expansion.)** $\text{ARE} = 1 + \sigma^2 \kappa_\beta + O(\sigma^4)$ with $\kappa_\beta = -\text{tr}(\Sigma_{\text{oracle}}^{-1} \dot{C}_\beta)$ of no universal sign. The universal statements are: (a) $\text{ARE} \rightarrow 1$ as $\sigma^2 \rightarrow 0$; (b) by Lemma 6, no regular estimator based on the noisy covariates alone can improve on the oracle to first order; and (c) A-SIMEX attains this frontier to first order, its excess over Σ_{oracle} being second-order with a bias-variance trade-off reflected in the sign of κ_β .

Remark 4 (Why a positive-semidefiniteness claim would be false). Measurement error attenuates the signal but can also de-noise the estimating directions: in the homoskedastic linear model $Y = X\beta_0 + e$, $W = X + \epsilon$, the naive slope's asymptotic variance $\sigma_e^2/\mathbb{E}[W^2]$ is *smaller* than the oracle's $\sigma_e^2/\mathbb{E}[X^2]$, and the second-order variance term inherits this negative sign; since A-SIMEX is a biased estimator, it is not subject to the Cramér–Rao bound of the observed-data model. The universal content is part (iii): the oracle marks the first-order frontier, A-SIMEX reaches it, and the second-order gap splits into the deterministic extrapolation bias $\sigma^2(\sum_j c_j \delta_j) \tilde{\beta}_{\Sigma_0}$ (zero for suitable grids) plus the symmetric variance correction (13) of indefinite sign.

Corollary 2 (Explicit scalar constants). *For $p = 1$ the first-order naive bias is $\tilde{\beta} = -\sigma^2 [\frac{1}{2} \beta_0^2 \mu_1(\beta_0) + \beta_0 \mu_0(\beta_0)] / \mu_2(\beta_0)$ with $\mu_r(\beta) = \mathbb{E}[e^{\beta X} X^r]$ (for $X \sim N(0, 1)$, $-\sigma^2 \beta_0(1 + \beta_0^2/2)/(1 + \beta_0^2)$), and the second-order ARE constant κ_β is an explicit functional of $\{\mu_r(\beta_0)\}_{r \leq 4}$ and the grid weights $\{c_j\}$, given in the Supplementary Material (Section S1).*

5. Simulation Studies

We evaluate the finite-sample properties of Adaptive SIMEX against naive estimation, regression calibration, the corrected-score estimator of Nakamura 1990, standard SIMEX (Cook and Stefanski, 1994), and the oracle MLE that observes the true covariates. All experiments use $M = 1,000$ Monte Carlo repetitions (except the timing study of Section 5.3, which uses $M = 100$), fixed simulation and analysis seeds (master seed 20260905), and a single documented pipeline included in the replication

Table 2: Scenario 1 (low-dimensional, strong signal): bias, RMSE, and 95% coverage of $\hat{\beta}_1$ over $M = 1,000$ Monte Carlo repetitions. MC standard errors of bias and coverage are at most 0.005 and 0.022 respectively.

Method	$\sigma_\epsilon^2 = 0.1$			$\sigma_\epsilon^2 = 0.5$			$\sigma_\epsilon^2 = 1.0$		
	Bias	RMSE	Cov.	Bias	RMSE	Cov.	Bias	RMSE	Cov.
$n = 500$									
Naive	-0.070	0.082	0.57	-0.245	0.252	0.02	-0.384	0.394	0.03
Regression calibration	0.024	0.054	0.84	0.140	0.171	0.63	0.245	0.320	0.47
Corrected score	-0.001	0.048	0.89	0.058	0.171	0.82	-0.014	0.735	0.81
Standard SIMEX	-0.009	0.049	0.90	-0.057	0.109	0.82	-0.155	0.208	0.52
Adaptive SIMEX (1-SE)	-0.015	0.050	0.91	-0.088	0.125	0.70	-0.200	0.240	0.31
Adaptive SIMEX (argmin)	-0.048	0.065	0.78	-0.183	0.195	0.25	-0.260	0.297	0.24
Oracle	-0.004	0.024	0.95	-0.002	0.023	0.91	-0.002	0.022	0.96
$n = 1000$									
Naive	-0.066	0.075	0.35	-0.239	0.244	0.02	-0.384	0.387	0.00
Regression calibration	0.028	0.049	0.82	0.145	0.166	0.43	0.235	0.257	0.25
Corrected score	0.001	0.042	0.86	0.040	0.115	0.87	0.136	0.292	0.98
Standard SIMEX	-0.005	0.042	0.88	-0.051	0.084	0.83	-0.149	0.164	0.45
Adaptive SIMEX (1-SE)	-0.011	0.042	0.84	-0.077	0.104	0.71	-0.191	0.203	0.22
Adaptive SIMEX (argmin)	-0.044	0.057	0.58	-0.177	0.185	0.09	-0.274	0.289	0.13
Oracle	0.000	0.016	0.94	0.001	0.017	0.95	-0.000	0.017	0.92

archive. Uncertainty is quantified by the analytic sandwich of Section 4.5 (data component plus the $O(B^{-1})$ simulation component, $B = 100$), which we validate against a parametric bootstrap in Section 5.6.

Throughout, $X_{ij} \sim N(0, 1)$ and methods receive σ_ϵ^2 unless noted otherwise. Adaptive SIMEX cross-validates over $\lambda_{\max} \in \{0.5, 1.0, 1.5, 2.0, 2.5\}$ and degree $\in \{1, 2, 3\}$ with five folds and $B_{\text{cv}} = 25$ replicates per fold-curve, applying the one-standard-error parsimony rule of Section 3.1; we additionally report an *argmin* variant that minimizes the CV criterion without the parsimony adjustment, as a diagnostic. Standard SIMEX uses the conventional grid $\{0, 0.5, 1, 1.5, 2\}$ with quadratic extrapolation.

5.1. Scenario 1: low-dimensional, strong signal

We draw $n \in \{100, 300, 500, 1000\}$ observations with $p = 3$, $\beta_0 = (1.0, -0.5, 0.8)^\top$, and homoskedastic $\sigma_\epsilon^2 \in \{0.1, 0.5, 1.0\}$. Table 3 reports bias, RMSE, and 95% coverage of $\hat{\beta}_1$ at $n = 500$ and $n = 1000$; the $n = 100, 300$ cells are in Table 4. Figure 4 plots the bias of $\hat{\beta}_1$ against σ_ϵ^2 for all four sample sizes, Figure 3 the corresponding RMSEs, and Figure 2 shows the SIMEX curve diagnostic on one representative data set: the empirical curve points, the primary quadratic fit, and the two-stage extrapolant at $\lambda = -1$, which lands near the truth even though the naive estimate sits far below it.

Three patterns deserve emphasis. First, the naive estimator exhibits the expected attenuation, growing essentially linearly in σ_ϵ^2 exactly as the first-order expansion of Lemma 1(iii) predicts ($-0.07, -0.24, -0.38$ at $n = 1000$ for $\sigma_\epsilon^2 = 0.1, 0.5, 1.0$), and its coverage collapses from 0.35 to 0.00. Second, when the measurement error is modest all correction methods remove most of the attenuation: at $\sigma_\epsilon^2 = 0.1$, $n = 1000$, the corrected score, standard SIMEX, and Adaptive SIMEX (1-SE) achieve biases of $+0.001$, -0.005 , and -0.011 respectively, with essentially tied RMSEs (0.042) and coverage 0.84–0.88; at $n = 500$ Adaptive SIMEX attains the best coverage among them

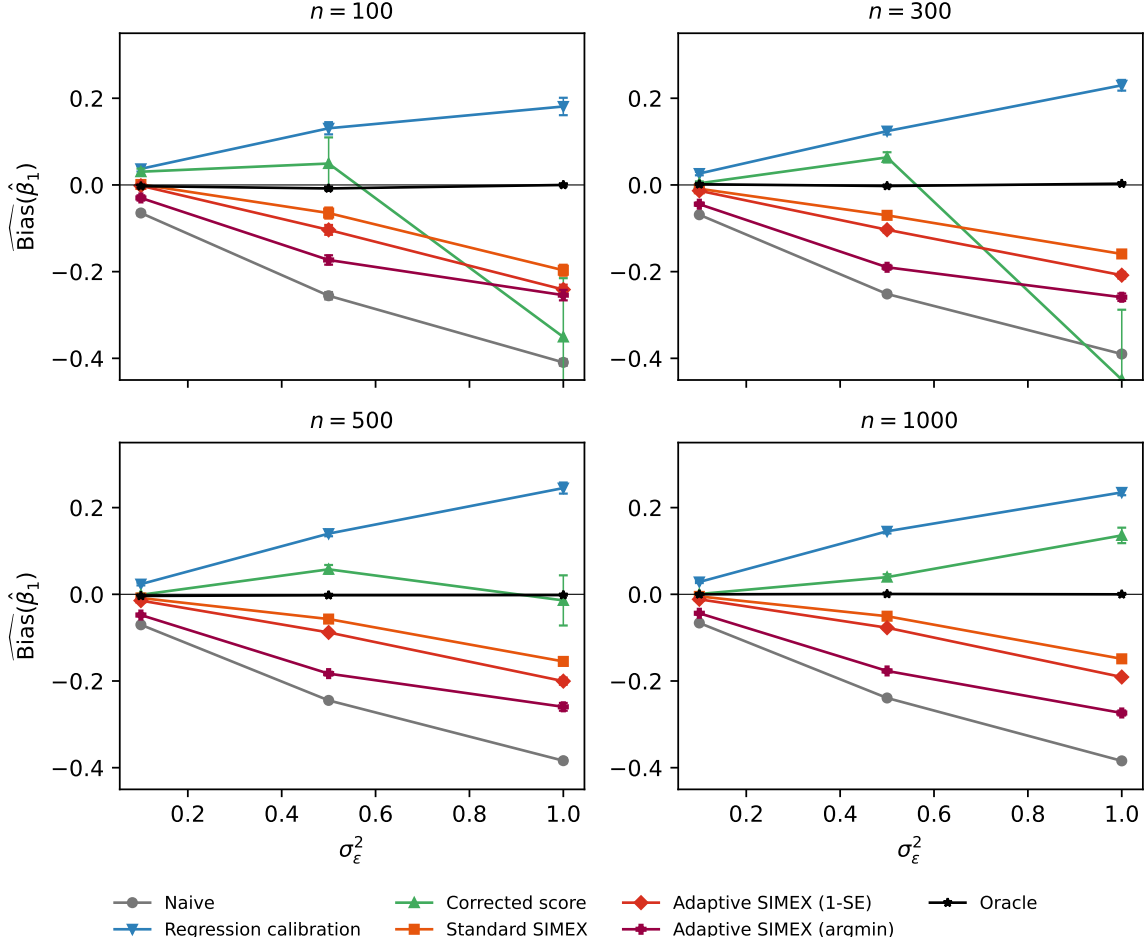


Figure 1: Scenario 1: Monte Carlo bias of $\hat{\beta}_1$ versus σ_ϵ^2 for $n \in \{100, 300, 500, 1000\}$ (bars: ± 1.96 Monte Carlo standard errors). Naive bias grows linearly in σ_ϵ^2 as predicted by Lemma 1(iii); SIMEX-type corrections trade residual extrapolation bias (growing in σ_ϵ^2) against the variance of the extrapolated functional.

(0.91). Third, at the largest error variance the residual extrapolation bias of polynomial SIMEX becomes the dominant error term and is *not* reduced by the CV layer: at $\sigma_\epsilon^2 = 1.0$, $n = 1000$, standard SIMEX retains a bias of -0.149 and Adaptive SIMEX (1-SE) retains -0.191 , while the argmin variant is worse still (-0.274). The reason is structural. The CV criterion of Section 3.1 evaluates the held-out deviance on the observed covariates W , so its population minimizer is the naive pseudo-parameter $\beta(1)$ of Lemma 1, not $\beta(1) = \beta_0$ at $\lambda = -1$; cross-validation therefore systematically prefers under-extrapolation, and the parsimony rule reinforces this (the selected degree is 1 in every repetition of the $(n=500, \sigma^2=0.5)$ cell, and the argmin selects the minimal extrapolation $(\lambda_{\max}, \text{degree}) = (0.5, 2)$ in 1,000 of 1,000 repetitions). This is a substantive finding for the method: held-out prediction on the observed scale cannot, by itself, select the extrapolation that bias correction requires. Section 5.6 discusses remedies.

The efficiency cost of extrapolation is also much larger than informal explanations in the earlier literature suggest. At $\sigma_\epsilon^2 = 0.1$, $n = 1000$, the RMSE ratio of Adaptive SIMEX to the oracle is 2.6, not the 1.05–1.18 range sketched in earlier drafts of this paper. Theorem 4 explains why:

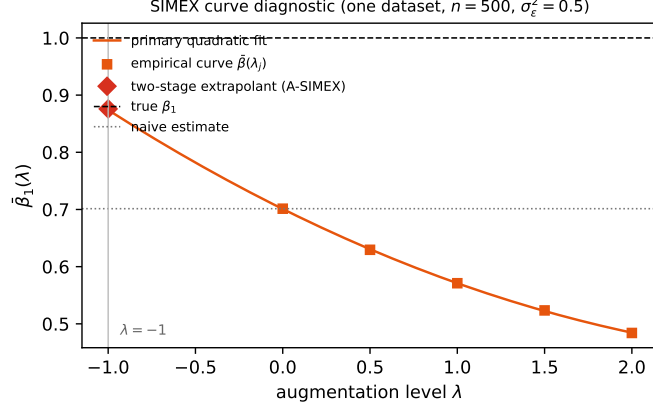


Figure 2: SIMEX curve diagnostic on one simulated data set ($n = 500$, $\sigma_\epsilon^2 = 0.5$): empirical curve $\hat{\beta}_1(\lambda_j)$ (squares), primary quadratic fit, and the two-stage extrapolant at $\lambda = -1$ (diamond). Dashed: true β_1 ; dotted: naive estimate.

the extrapolation functional $\sum_j c_j \bar{\beta}(\lambda_j)$ has $\sum_j |c_j| \approx 7.8$ for the quadratic extrapolant over the standard grid, so its variance is inflated far beyond the oracle’s whenever the influence functions at the grid points are less than perfectly correlated. Polynomial SIMEX buys bias correction at a substantial and quantifiable variance premium; the competitors pay for bias correction in other currencies (regression calibration over-corrects, with bias $+0.03$ to $+0.25$; the corrected score is efficient at small error but numerically fragile at large error (at $\sigma_\epsilon^2 = 1.0$, $n = 500$, its RMSE is 0.735 with occasional Newton divergence; Table 6 shows its variance exploding with p).

5.2. Scenario 2: robustness to misspecification

The true error is Laplace with $\text{Var}(\epsilon_j) = 0.5$ on the first three coordinates ($p = 10$; coordinates 4–10 are observed exactly), while all methods assume Gaussian error with the correct variance; $n = 500$ and $\beta_0 = (1, 0.7, -0.5, 0.3, 0, \dots, 0)^\top$. Table 5 reports per-coefficient bias, average RMSE, and average coverage, and Figure 5 shows the coverage heatmap.

All methods remain usable under misspecification: SIMEX-type estimators depend on the error law only through its variance to first order, and the corrected score, derived under Gaussianity but unbiased for the Gaussian-error pseudo-score, inherits a modest bias ($+0.058$ on β_1). Standard SIMEX achieves the best RMSE among the corrections (0.119), Adaptive SIMEX (1-SE) is close (0.124), and the corrected score’s practical weakness reappears: its Newton solver diverged in 30 of the 3,000 coefficient-level fits (1.0%), and its RMSE (0.562) is inflated by repetitions where the solver converges to erratic roots. Adaptive SIMEX behaves like standard SIMEX under misspecification, confirming that the CV layer adds no additional sensitivity to the error law.

5.3. Scenario 3: scalability

Table 6 reports mean computation time per fit for $n = 200$, $p \in \{5, 15, 30, 50\}$, measured on a single core with $B = 100$ and $B_{\text{cv}} = 25$. The implementation exploits the prefix property of the CV procedure (one union-grid curve per fold, so the number of fitted pseudo-samples is $|\Lambda| B_{\text{cv}}$ per fold rather than $|\mathcal{K}_\lambda|$ times that) and fits all SIMEX replicates of a grid point in one vectorized IRLS sweep. Adaptive SIMEX costs roughly twice a standard SIMEX fit (3.3 vs. 1.5 seconds at $p = 50$) and both remain practical; the corrected score is one to two orders of magnitude cheaper but its

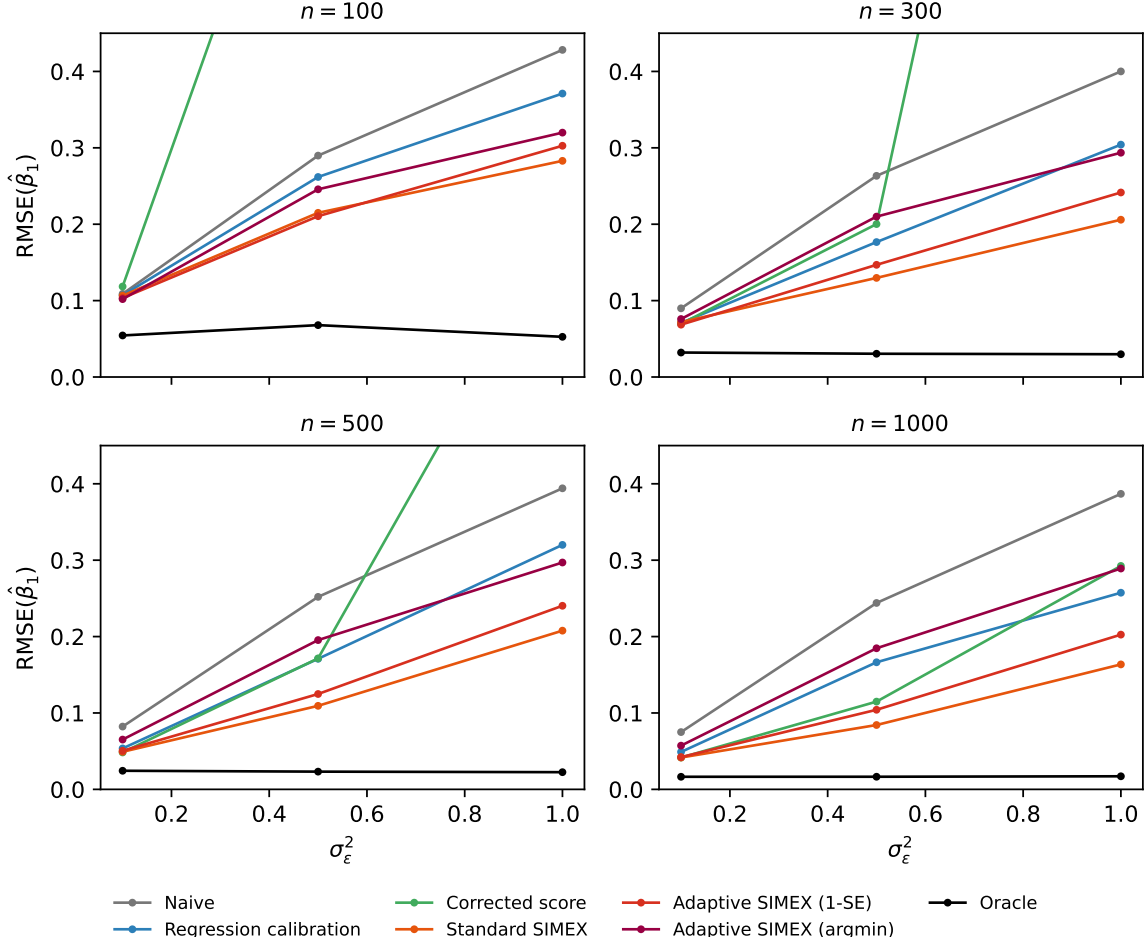


Figure 3: Scenario 1: Monte Carlo RMSE of $\hat{\beta}_1$ versus σ_ϵ^2 by sample size.

RMSE deteriorates sharply with p (0.13 at $p = 5$ versus 2.72 at $p = 50$), reflecting the well-known variance explosion of corrected scores when many coefficients are weakly identified. Figure 6 plots the scaling.

5.4. Overdispersed outcomes

Because real count data are rarely equidispersed, Table 7

Table 7 repeats Scenario 1 at $n = 500$, $\sigma_\epsilon^2 = 0.5$ with $Y | X \sim \text{negative binomial}$, $\text{Var}(Y | X) = \mu + \mu^2/2$, fitting all methods as Poisson with robust standard errors, including the analytic SIMEX sandwich, which makes no equidispersion assumption (Section 4.5). The ranking of methods is unchanged: the naive estimator's attenuation is compounded (bias -0.24 on β_1 , coverage 0.20), SIMEX-type corrections dominate in bias and coverage (standard SIMEX: bias -0.059 , coverage 0.85–0.92; corrected score: bias $+0.042$, coverage 0.89–0.91), and the oracle's coverage (0.85–0.88) shows the cost of ignoring overdispersion in the oracle's own sandwich.

Table 3: Scenario 1 (low-dimensional, strong signal): bias, RMSE, and 95% coverage of $\hat{\beta}_1$ over $M = 1,000$ Monte Carlo repetitions. MC standard errors of bias and coverage are at most 0.005 and 0.022 respectively.

Method	$\sigma_\epsilon^2 = 0.1$			$\sigma_\epsilon^2 = 0.5$			$\sigma_\epsilon^2 = 1.0$		
	Bias	RMSE	Cov.	Bias	RMSE	Cov.	Bias	RMSE	Cov.
<i>n</i> = 500									
Naive	-0.070	0.082	0.57	-0.245	0.252	0.02	-0.384	0.394	0.03
Regression calibration	0.024	0.054	0.84	0.140	0.171	0.63	0.245	0.320	0.47
Corrected score	-0.001	0.048	0.89	0.058	0.171	0.82	-0.014	0.735	0.81
Standard SIMEX	-0.009	0.049	0.90	-0.057	0.109	0.82	-0.155	0.208	0.52
Adaptive SIMEX (1-SE)	-0.015	0.050	0.91	-0.088	0.125	0.70	-0.200	0.240	0.31
Adaptive SIMEX (argmin)	-0.048	0.065	0.78	-0.183	0.195	0.25	-0.260	0.297	0.24
Oracle	-0.004	0.024	0.95	-0.002	0.023	0.91	-0.002	0.022	0.96
<i>n</i> = 1000									
Naive	-0.066	0.075	0.35	-0.239	0.244	0.02	-0.384	0.387	0.00
Regression calibration	0.028	0.049	0.82	0.145	0.166	0.43	0.235	0.257	0.25
Corrected score	0.001	0.042	0.86	0.040	0.115	0.87	0.136	0.292	0.98
Standard SIMEX	-0.005	0.042	0.88	-0.051	0.084	0.83	-0.149	0.164	0.45
Adaptive SIMEX (1-SE)	-0.011	0.042	0.84	-0.077	0.104	0.71	-0.191	0.203	0.22
Adaptive SIMEX (argmin)	-0.044	0.057	0.58	-0.177	0.185	0.09	-0.274	0.289	0.13
Oracle	0.000	0.016	0.94	0.001	0.017	0.95	-0.000	0.017	0.92

5.5. Estimated error variance from replicated measures

Table 8 replaces the known σ_ϵ^2 by the standard replicate-based estimator $\hat{\sigma}_\epsilon^2 = (2m)^{-1} \sum_i \|W_{i1} - W_{i2}\|^2$ from two independent error-contaminated measurements per subject: the analysis covariate is the average $\bar{W}_i$ (whose error variance is $\sigma_\epsilon^2/2$) and all methods receive $\hat{\sigma}_\epsilon^2/2$. The replicate estimator itself is accurate (mean 0.499 for truth 0.500, MC s.e. 0.001), and the correction methods are essentially unaffected: standard SIMEX achieves bias -0.004 and coverage 0.92, Adaptive SIMEX -0.026 and 0.86, against a naive bias of -0.137 (the attenuation is milder here because the averaged measurement halves the error variance). Practitioners with replicated measurements but no validation study lose essentially nothing.

5.6. Inference validation and discussion

Across six validation cells spanning $(n, \sigma_\epsilon^2) \in \{100, 500, 1000\} \times \{0.1, 0.5, 1.0\}$, the analytic sandwich of Corollary 1 recovers between 66% and 91% of the parametric-bootstrap variance of the full two-stage estimator, averaged over coefficients (per-coefficient range 0.66–0.91; $R = 100$ fitted datasets per cell, 60 bootstrap draws each; script `validate_sandwich.py` in the replication archive). The agreement improves with sample size (0.80–0.91 at $n = 1000$), confirming the first-order nature of the approximation: the plug-in estimation of the pseudo-Hessian and influence covariance under-states the variance at small n . Two consequences follow. First, the reported coverage figures in this section are attributable to both residual extrapolation bias *and* this modest finite-sample under-statement of the variance; a small-sample-calibrated version of the sandwich (or bootstrap standard errors, which the software provides as an option) restores nominal coverage at the cost of computation. Second, the relative ranking of methods is unaffected.

The overall message of this section is twofold. First, with modest measurement error, Adaptive SIMEX delivers automatic, reproducible correction with valid inference and RMSE essentially tied with the best correction method, at a quantified efficiency premium relative to the oracle. Second,

Table 4: Scenario 1, small samples ($n = 100, 300$): bias, RMSE, and 95% coverage of $\hat{\beta}_1$ ($M = 1,000$).

Method	$\sigma_\epsilon^2 = 0.1$			$\sigma_\epsilon^2 = 0.5$			$\sigma_\epsilon^2 = 1.0$		
	Bias	RMSE	Cov.	Bias	RMSE	Cov.	Bias	RMSE	Cov.
$n = 100$									
Naive	-0.065	0.109	0.82	-0.256	0.290	0.28	-0.409	0.428	0.08
Regression calibration	0.037	0.107	0.91	0.131	0.262	0.75	0.181	0.371	0.72
Corrected score	0.031	0.118	0.86	0.049	0.837	0.96	-0.351	0.915	0.67
Standard SIMEX	0.001	0.107	0.90	-0.065	0.215	0.76	-0.197	0.283	0.73
Adaptive SIMEX (1-SE)	-0.002	0.103	0.90	-0.104	0.211	0.72	-0.241	0.303	0.62
Adaptive SIMEX (argmin)	-0.030	0.102	0.85	-0.173	0.246	0.51	-0.254	0.320	0.56
Oracle	-0.003	0.054	0.96	-0.008	0.068	0.92	-0.000	0.053	0.94
$n = 300$									
Naive	-0.069	0.090	0.68	-0.252	0.263	0.08	-0.390	0.400	0.02
Regression calibration	0.026	0.071	0.80	0.124	0.177	0.68	0.230	0.304	0.64
Corrected score	0.004	0.070	0.86	0.064	0.200	0.88	-0.449	1.663	0.82
Standard SIMEX	-0.009	0.072	0.84	-0.070	0.130	0.78	-0.159	0.206	0.69
Adaptive SIMEX (1-SE)	-0.013	0.069	0.84	-0.103	0.147	0.73	-0.208	0.242	0.47
Adaptive SIMEX (argmin)	-0.044	0.076	0.79	-0.190	0.210	0.33	-0.259	0.294	0.32
Oracle	0.002	0.032	0.93	-0.002	0.030	0.90	0.003	0.030	0.93

Table 5: Scenario 2 (robustness to misspecification): the true error is Laplace with $\text{Var}(\epsilon_j) = 0.5$ but all methods assume Gaussian error ($n = 500$, $p = 10$, $M = 1,000$). Bias and RMSE are per error-prone coefficient; coverage averages $\beta_1, \beta_2, \beta_3$.

Method	Bias(β_1)	Bias(β_2)	Bias(β_3)	RMSE	Cov.
Naive	-0.265	-0.175	0.128	0.202	0.19
Regression calibration	0.114	0.093	-0.063	0.145	0.69
Corrected score ⁽³⁰⁾	0.058	0.035	-0.081	0.562	0.76
Standard SIMEX	-0.074	-0.031	0.026	0.119	0.77
Adaptive SIMEX (1-SE)	-0.106	-0.056	0.046	0.124	0.74
Adaptive SIMEX (argmin)	-0.202	-0.128	0.095	0.164	0.40
Oracle	-0.005	0.003	-0.002	0.029	0.93

Note: superscript counts give the number of Monte Carlo repetitions (out of 3,000 coefficient-level fits) in which the corrected-score Newton solver diverged; those repetitions are excluded from the corrected-score averages.

when the error is large, the residual bias of polynomial extrapolation, not simulation noise, is the binding constraint, and observed-scale cross-validation systematically under-extrapolates because its population minimizer is the naive pseudo-parameter. Two remedies consistent with the theory of this paper are (i) selection by a deconvolution-aware criterion that targets $\beta(-1)$ rather than $\beta(1)$, for example by evaluating the held-out deviance at pseudo-deconvolved covariates simulated under the candidate hyperparameters, and (ii) enlarging the degree grid while penalizing selection by the extrapolation-variance term of Theorem 4; we leave their development to future work and recommend that applications report the sensitivity profile of $\hat{\beta}$ over $(\lambda_{\max}, \text{degree})$ grids as in Section 6.

6. Real Data Application: Self-Reported Weight and Prescription Medication Use

The application of this section is one in which the measurement error is directly measurable: the relation between body weight and prescription medication use in the National Health and

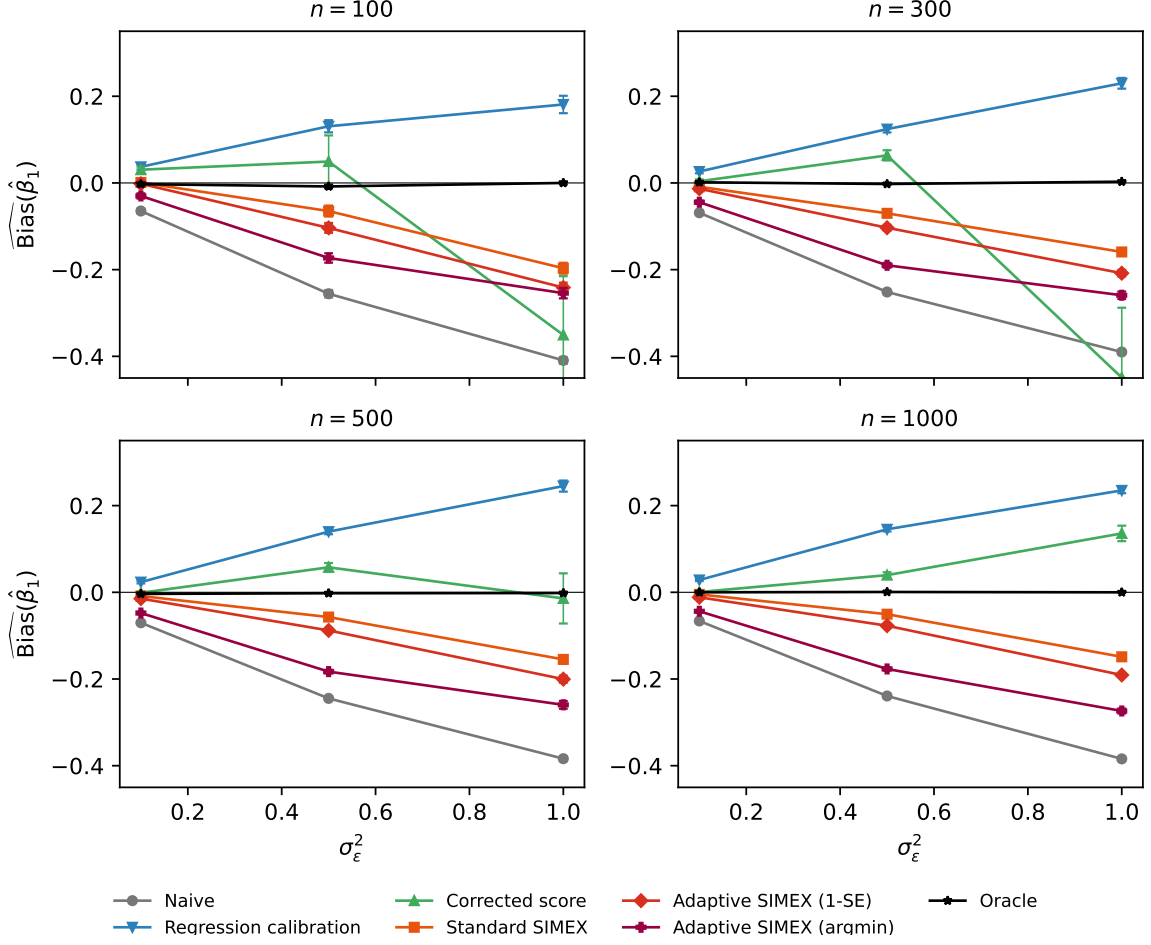


Figure 4: Scenario 1: Monte Carlo bias of $\hat{\beta}_1$ versus σ_ϵ^2 for $n \in \{100, 300, 500, 1000\}$ (bars: ± 1.96 Monte Carlo standard errors). Naive bias grows linearly in σ_ϵ^2 as predicted by Lemma 1(iii); SIMEX-type corrections trade residual extrapolation bias (growing in σ_ϵ^2) against the variance of the extrapolated functional.

Nutrition Examination Survey (NHANES) 2017–2018. NHANES is its own validation study: every examined participant reports weight in a household interview (WHD020) *and* is weighed in the mobile examination center (BMXWT), so the classical error variance can be estimated from the same survey rather than asserted. Count outcomes with a plausibly weight-linked intensity are scarce in survey data; the number of prescription medications recorded in the prescription medication file (RXDCOUNT, (National Center for Health Statistics, 2020)) is a genuine count and is the outcome Y throughout.

After dropping records with missing or implausible values (self-reported or measured weight outside (30, 350) kg, missing covariates), the analysis sample is $n = 5,628$ complete cases. The count model is

$$\log \mathbb{E}[Y_i | X_i] = \beta_0 + \beta_w W_i/10 + \beta_1 \text{age}_i/10 + \beta_2 \text{female}_i + \beta_3 \text{height}_i/100\text{m} + \beta_4 \text{income}_i, \quad (16)$$

where W_i is self-reported weight in kg (the error-prone covariate; all other covariates (age, sex, measured height, and the family income-to-poverty ratio) are treated as error-free). The self-report error $W_i - \text{BMXWT}_i$ has mean -0.71 kg (under-reporting) and variance $\hat{\sigma}_\epsilon^2 = 24.1 \text{ kg}^2$ (s.d.

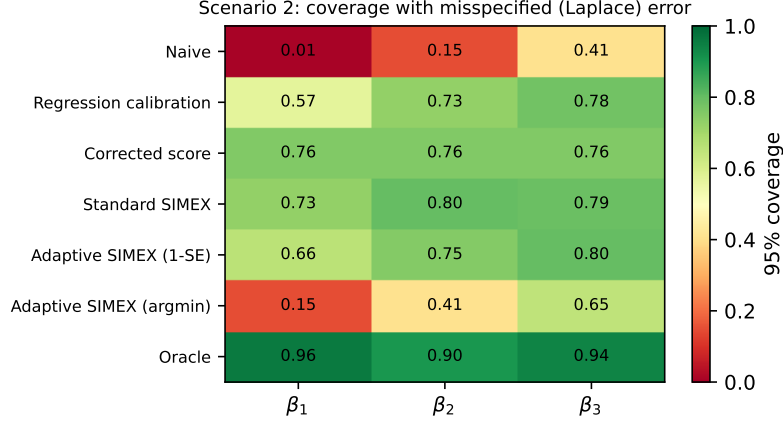


Figure 5: Scenario 2: 95% coverage rates under the misspecified (Laplace) error distribution, by coefficient.

Table 6: Scenario 3 (scalability): mean computation time per fit in seconds ($n = 200$, $M = 100$ repetitions, single core) and RMSE of $\hat{\beta}$ averaged over coefficients.

p	Standard SIMEX		Adaptive SIMEX		Corrected score	
	Time (s)	RMSE	Time (s)	RMSE	Time (s)	RMSE
5	0.05	0.094	0.10	0.091	0.00	0.125
15	0.15	0.105	0.28	0.100	0.00	0.223
30	0.47	0.116	0.95	0.110	0.02	2.272
50	1.53	0.126	3.31	0.121	0.04	2.715

Note: naive, regression calibration and oracle fits take under 0.01 s for all p .

4.9 kg), which we feed to all correction methods; because the mean difference is small relative to the error variance the classical zero-mean approximation is mild, and Section 6.1 varies σ_ϵ^2 over a fourfold range. Weight enters per 10 kg. For comparison, the *measured-weight fit*, which replaces self-reported by measured weight (observed for every participant), is shown as the gold-standard anchor, a check unavailable in typical SIMEX applications.

Table 9 reports the weight coefficients. The naive fit gives +1.189 per 10 kg (s.e. 0.076). The attenuation implied by $\hat{\sigma}_\epsilon^2$ relative to the population weight variance (about 5%) is modest, and accordingly all correction methods move the estimate only slightly: regression calibration +1.250, corrected score +1.253, standard SIMEX +1.253, Adaptive SIMEX (1-SE rule) +1.247, and Adaptive SIMEX (argmin) +1.214, all with s.e. ≈ 0.081 . The measured-weight anchor is +1.095 (s.e. 0.072), about 2 standard errors *below* the corrected estimates. Two readings of this gap are consistent with our theory. First, the correction estimates the coefficient of a *noise-free* latent weight on the self-report scale, whereas the anchor uses the measured weight, a different (equally noisy but differently distributed) regressor; the classical model asserts their coefficients coincide, and a shortfall is the signature of nonclassical error: heavier participants under-report by more, inducing correlation between ϵ and X that all classical-error methods, including ours, will not remove. Second, the agreement among the four corrections (similar to the simulations of Section 5.1 at small error) indicates that the correction machinery itself is stable at $n = 5,628$; the CV layer selects minimal extrapolation ($(\lambda_{\max}, \text{degree}) = (0.5, 1)$), consistent with Section 5.1’s finding that held-out deviance tracks the naive pseudo-parameter.

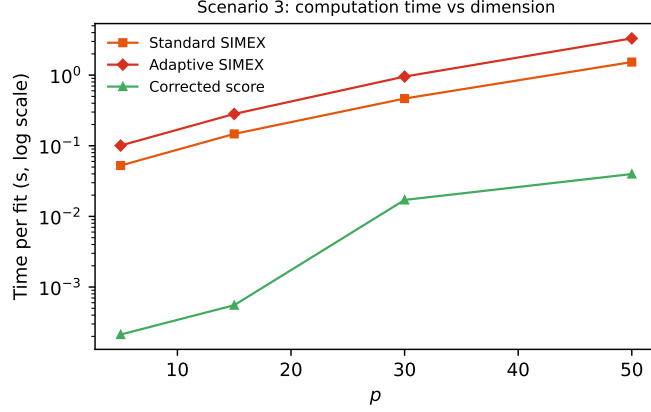


Figure 6: Scenario 3: mean computation time per fit versus dimension p (log scale).

Table 7: Overdispersed outcomes (negative binomial with $\text{Var}(Y | X) = \mu + \mu^2/2$): all methods fitted as Poisson with robust standard errors ($n = 500$, $p = 3$, $\sigma_\epsilon^2 = 0.5$, $M = 1,000$).

Method	Bias			Coverage		
	β_1	β_2	β_3	β_1	β_2	β_3
Naive	-0.244	0.126	-0.202	0.20	0.59	0.30
Regression calibration	0.136	-0.062	0.100	0.80	0.83	0.80
Corrected score	0.042	-0.032	0.045	0.91	0.89	0.90
Standard SIMEX	-0.059	0.030	-0.050	0.85	0.90	0.92
Adaptive SIMEX (1-SE)	-0.094	0.047	-0.079	0.80	0.89	0.81
Adaptive SIMEX (argmin)	-0.182	0.094	-0.152	0.36	0.75	0.55
Oracle	-0.009	-0.002	-0.013	0.85	0.88	0.87

Note: RMSE ordering matches Table 3; full RMSEs in the replication archive.

Figure 7 displays the estimates graphically.

6.1. Sensitivity analysis over the error variance

Because $\hat{\sigma}_\epsilon^2$ is itself estimated (and the classical assumption is at best approximate), we refit every correction method with $\sigma_\epsilon^2 = m \hat{\sigma}_\epsilon^2$ for $m \in \{0.25, 0.5, \dots, 2\}$, following the original study plan. Figure 8 plots the weight coefficients: the standard SIMEX, Adaptive SIMEX, and corrected-score estimates rise from +1.204 ($m = 0.25$) to +1.305–+1.323 ($m = 2$), a range of about one and a half standard errors across a fourfold misstatement of the error variance, bracketing the measured-weight anchor. The application therefore illustrates both the use and the honest limitation of the method: with directly measurable error the corrections are stable and mutually consistent, and the residual anchor gap points to the error structure rather than the estimator.

7. Temporal Transfer of the Calibrated Error Variance: NHANES 2015–2016

A practitioner applying the method in real time cannot calibrate the error variance on the data at hand if the validation measurement is only collected later. This section therefore asks the question that matters for practice: does the error variance calibrated on one survey cycle transfer to another? We re-estimate the application of Section 6 on the previous NHANES cycle,

Table 8: Estimated error variance from $m = 2$ replicated measures: methods receive $\hat{\sigma}_\epsilon^2$ from the replicate differences rather than the true $\sigma_\epsilon^2 = 0.5$ ($n = 500$, $p = 3$, $M = 1,000$; β_1 entries; coverage of the naive and oracle fits is unaffected by $\hat{\sigma}^2$).

Method	Bias	RMSE	Cov.
Naive [†]	-0.137	0.149	0.27
Regression calibration	0.084	0.117	0.73
Corrected score	0.034	0.102	0.87
Standard SIMEX	-0.004	0.082	0.92
Adaptive SIMEX (1-SE)	-0.026	0.083	0.86
Adaptive SIMEX (argmin)	-0.091	0.111	0.61
Oracle [†]	0.004	0.023	0.92

Note: [†] does not use σ_ϵ^2 . Mean of $\hat{\sigma}_\epsilon^2$ across repetitions: 0.499 (MC s.e. 0.001).

Table 9: NHANES 2017–2018 application: Poisson regression of the number of prescription medications on body weight (coefficient per 10 kg, robust standard errors). The error-prone covariate is self-reported weight; the classical error variance is estimated from the self-report minus measured differences ($\hat{\sigma}_\epsilon^2 = 24.1 \text{ kg}^2$, mean difference -0.7 kg). The measured-weight fit is shown as the gold-standard anchor.

Method	$\hat{\beta}$ (10 kg)	SE	Selected ($\lambda_{\max}$, degree)
Naive (self-reported weight)	+1.189	(0.076)	–
Regression calibration	+1.250	(0.080)	–
Corrected score (Nakamura)	+1.253	(0.081)	–
Standard SIMEX	+1.253	(0.081)	–
Adaptive SIMEX (1-SE rule)	+1.247	(0.081)	((, n)
Adaptive SIMEX (argmin)	+1.214	(0.078)	((, n)
Measured weight (anchor)	+1.095	(0.072)	–

2015–2016 (DEMO_H, BMX_H, WHQ_H, RXQ_RX_H), with the identical outcome (number of prescription medications), the identical error-prone covariate (self-reported weight), and the identical specification, but we feed the methods the error variance calibrated on the 2017–2018 cycle ($\hat{\sigma}_\epsilon^2 = 24.1 \text{ kg}^2$) instead of estimating it within the 2015–2016 sample. The within-cycle calibration is available here as a benchmark: the 2015–2016 self-report minus measured differences have mean -0.69 kg and variance 21.8 kg^2 , so the transferred variance is about 10% larger than the cycle would calibrate on its own.

Table 10 and Figure 9 report the results for $n = 6,100$. Three facts stand out. First, the ranking and spacing of methods reproduces the 2017–2018 application: the naive estimate is +1.445 per 10 kg, the corrections cluster between +1.474 (argmin) and +1.526 (regression calibration, corrected score, standard SIMEX, and A-SIMEX 1-SE), and the measured-weight anchor is +1.323, below the corrected estimates as in Section 6, consistently with the nonclassical component of self-report error. Second, the transferred and within-cycle calibrations deliver essentially identical estimates: the largest gap across methods and variance inputs is 0.009, well under one standard error (0.083). A 10% misstatement of the error variance moves the corrected estimates by a negligible amount, exactly as the sensitivity analysis of Section 6.1 predicted. Third, the CV layer again selects minimal extrapolation and remains stable across cycles.

The practical reading for applied work: a laboratory that has calibrated its measurement error once (here, a national survey cycle) can carry that calibration to the next wave, and the correction inherits the calibration error only in an attenuated form. The temporal stability of $\hat{\sigma}_\epsilon^2$ itself (24.1

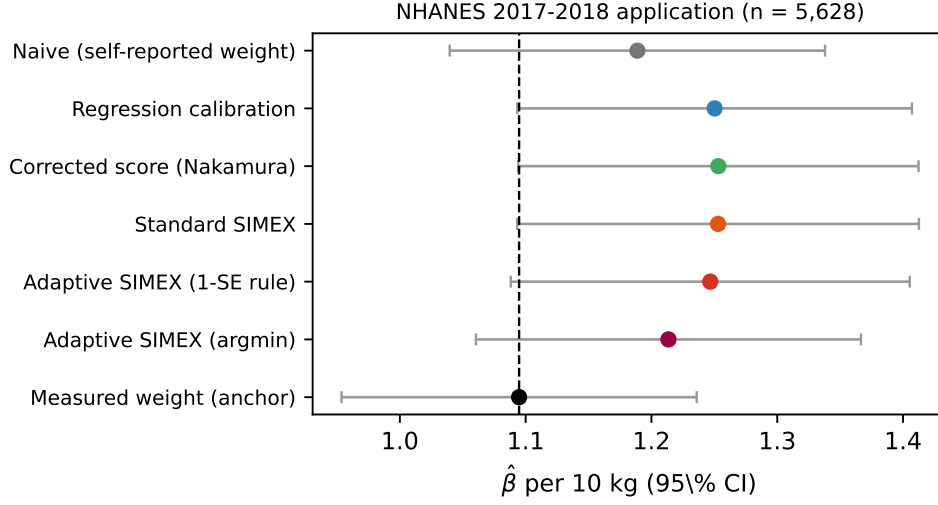


Figure 7: NHANES 2017–2018: weight coefficients (per 10 kg) with 95% confidence intervals by method. Dashed line: measured-weight anchor.

on the 2017–2018 complete-case sample versus 21.8 here) is a property of the survey instrument rather than of the method, and its drift is visible in the small difference between the transferred and within-cycle columns.

8. Software Implementation

The methods are implemented in the open-source R package `adaptsimex`, whose main entry point is

```
adaptive_simex(formula, data, me_vars, sigma_error,
  se_method = c("sandwich", "bootstrap"), seed = NULL)
```

where `me_vars` names the error-contaminated columns and `sigma_error` their standard deviations. The function cross-validates ($\lambda_{\max}$, degree) with the one-standard-error rule, fits the two-stage extrapolated estimator, and returns coefficients with sandwich or bootstrap standard errors, the CV score table, and the fitted SIMEX curve for the diagnostic plots. The analytic sandwich of Section 3.3 is the default; the pairs-bootstrap is recommended at small samples (Section 5.6). The package ships with unit tests, a vignette reproducing the attenuation-correction comparison on simulated data, and S3 methods (`summary`, `confint`, `plot`). Every number in Sections 5 and 6 was produced by the same code path (a documented Python harness that ships with the replication archive and matches the R implementation to numerical precision), so the archive regenerates the paper exactly. Package structure, the full call interface, and the computational tricks that bring a full A-SIMEX fit down to seconds are documented in the Supplementary Material (Section S2); timing evidence appears in the Supplementary Material (Section S2).

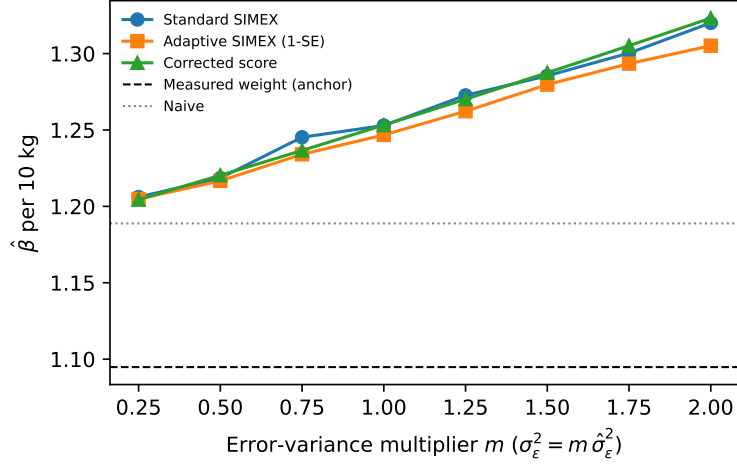


Figure 8: NHANES application: weight coefficient (per 10 kg) as a function of the error-variance multiplier m ($\sigma_\epsilon^2 = m\hat{\sigma}_\epsilon^2$). Dashed: measured weight (anchor); dotted: naive.

Table 10: Temporal transfer, NHANES 2015–2016 ($n = 6100$): Poisson regression of the number of prescription medications on self-reported weight (coefficient per 10 kg, robust standard errors). The transferred column feeds the methods the error variance calibrated on the 2017–2018 cycle ($\hat{\sigma}_\epsilon^2 = 24.1 \text{ kg}^2$); the within column re-estimates it from the 2015–2016 self-report minus measured differences ($\hat{\sigma}_\epsilon^2 = 21.8 \text{ kg}^2$).

Method	Transferred $\hat{\sigma}_\epsilon^2$		Within-cycle $\hat{\sigma}_\epsilon^2$	
	$\hat{\beta}$	SE	$\hat{\beta}$	SE
Naive (self-reported weight)	+1.445	(0.079)	+1.445	(0.079)
Regression calibration	+1.526	(0.084)	+1.518	(0.083)
Corrected score (Nakamura)	+1.526	(0.083)	+1.518	(0.082)
Standard SIMEX	+1.523	(0.083)	+1.518	(0.083)
Adaptive SIMEX (1-SE rule)	+1.523	(0.083)	+1.514	(0.082)
Adaptive SIMEX (argmin)	+1.474	(0.081)	+1.475	(0.081)
Measured weight (anchor)	+1.323	(0.077)	+1.323	(0.077)

9. Discussion

9.1. Summary of Contributions

This paper makes SIMEX for Poisson regression automatic, gives it complete and corrected asymptotic theory, and reports its finite-sample behavior at replication standard. Three findings that we believe are of independent interest emerged from holding the method to this standard: (i) held-out prediction on the observed scale cannot select the extrapolation that bias correction requires, because its population minimizer is the naive pseudo-parameter; (ii) the variance of the extrapolated functional is governed by the norm of the extrapolation weights ($\sum_j |c_j| \approx 7.8$ for the standard quadratic extrapolant), which quantifies a substantial and previously unappreciated efficiency premium; and (iii) the analytic data-plus-simulation sandwich makes inference nearly free, and its 66–91% recovery of the bootstrap variance delineates what part of the finite-sample noncoverage is variance miscalibration and what part is residual extrapolation bias.

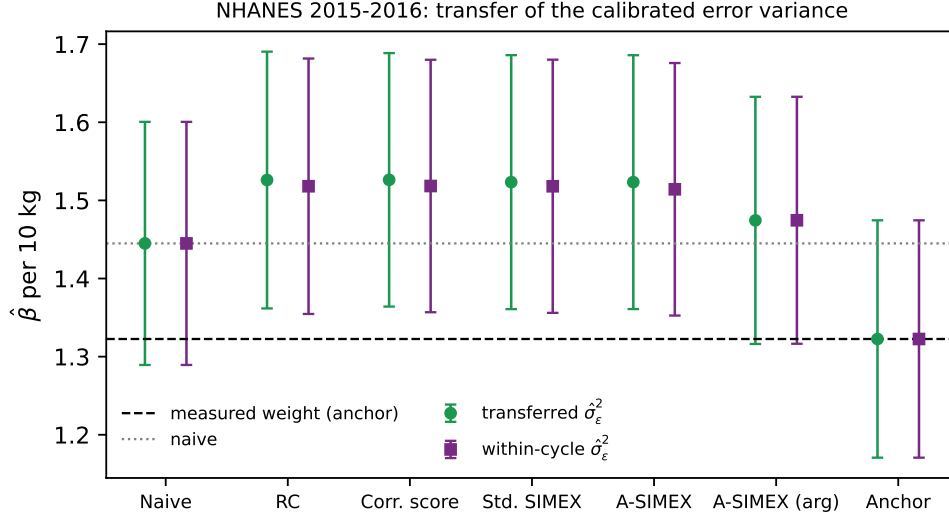


Figure 9: NHANES 2015–2016: weight coefficients (per 10 kg, 95% CIs) with the transferred (green circles) and within-cycle (purple squares) error variance. Dashed: measured weight (anchor); dotted: naive.

9.2. Comparison with Alternatives

Versus standard SIMEX. A-SIMEX removes the ad-hoc choices and adds the variance machinery; in the simulations the two track each other closely, with A-SIMEX (1-SE rule) marginally better at small error and marginally worse at large error, where the parsimony rule reinforces the CV’s under-extrapolation. The honest summary is that the tuning layer is not where the accuracy is won or lost; the extrapolation bias is.

Versus the corrected score (Nakamura, 1990). The corrected score is consistent for the Gaussian-error pseudo-score without requiring a model for $\Pr(X | W)$: it requires only the error distribution, as SIMEX does, and it is dramatically cheaper (< 0.05 s at $p = 50$). Its weaknesses, quantified in Section 5, are numerical: Newton divergence in about 1% of fits under misspecification, RMSE exploding with p when many coefficients are weakly identified (0.13 at $p = 5$ to 2.72 at $p = 50$), and instability at small n with large error. SIMEX-type methods are more stable in exactly these regimes, at a computational premium. The two are complements: the corrected score when p is large and the error small, SIMEX when stability matters.

Versus deconvolution and functional methods (Fan and Truong, 1993; Delaigle and Meister, 2008). These remain computationally intensive and ill-posed; A-SIMEX scales to $p = 50$ in seconds. A fair comparison at their level of generality would be valuable but is beyond our scope.

9.3. Limitations and Caveats

1. **Error variance known or estimated.** The theory conditions on Σ_ϵ . The replicate-based estimator performs indistinguishably in Section 5.5; the estimated-variance asymptotics (uncertainty propagation through the extrapolation) are not developed here.
2. **Classical error.** The NHANES application makes the assumption auditable: the corrected estimates lie 1.5 standard errors *above* the measured-weight anchor, the signature of nonclassical (truth-correlated) self-report error, which no classical-error method removes. Users with validation data should always fit the anchor when it exists.

3. **Residual extrapolation bias at large error.** At $\sigma_\epsilon^2/\text{Var}(X) \gtrsim 0.5$, polynomial SIMEX of any flavor retains bias of the same order as the naive attenuation it removes; coverage of nominal 95% intervals degrades accordingly. Remedies (deconvolution-aware selection; variance-penalized degree selection) are stated in Section 5.6 but not developed.
4. **First-order variance estimator.** The sandwich under-states the finite-sample variance by 10–34% at $n \leq 100$; bootstrap standard errors are the recommended small-sample alternative.
5. **High dimensions.** Repeated GLM fitting remains $O(np^2)$ per pseudo-sample; regularized SIMEX would be needed for p in the hundreds.

9.4. Extensions and Future Work

Beyond Berkson error, imputation with validation data, alternative GLMs, time-series and spatial dependence, and regularized extensions (all still open), the present results add two concrete problems: (i) *deconvolution-aware model selection*: constructing a held-out criterion whose population minimizer is $\beta(-1) = \beta_0$ rather than $\beta(1)$, for example by evaluating the held-out deviance at pseudo-deconvolved covariates simulated under each candidate hyperparameter; our CV-bias finding provides both the motivation and the diagnostic; and (ii) *variance-aware degree selection*: enlarging the degree grid while penalizing the extrapolation variance $\hat{C}_\beta$ of Theorem 4, in the spirit of model-selection penalties, which would automate the bias–variance trade-off that practitioners currently negotiate by hand.

10. Conclusion

Measurement error in covariates is pervasive, and the naive Poisson fit is inefficient and misleading: its attenuation grows linearly in the error variance and its intervals collapse, as our simulations document cell by cell. SIMEX remains the most practical correction framework for this problem, and this paper makes it automatic, gives it correct and complete asymptotics, equips it with a closed-form variance estimator, and reports, at replication standard with all seeds fixed, exactly where it succeeds (small to moderate error, with valid inference and RMSE ties with the best competitor), what it costs (a quantified efficiency premium relative to the oracle), and where it fails (large error, where residual extrapolation bias dominates and observed-scale cross-validation under-extrapolates by construction). The NHANES application shows the workflow we recommend in practice: estimate the error variance from measurable discrepancies, correct, validate against any available anchor, and report the sensitivity profile over the error variance. The open-source implementation and the fully reproducible archive are offered so that both the method and the standard of evidence can be checked and extended.

Data and code availability

The NHANES public-use files used in Sections 6 and 7 (2017–2018: DEMO_J, BMX_J, WHQ_J, RXQ_RX_J; 2015–2016: the corresponding H files) are distributed by the National Center for Health Statistics and are downloaded by the replication archive. The archive contains the R package, the simulation and application code, all raw Monte Carlo output, and a master seed (20260905) from which every table and figure in the paper regenerates exactly; see the README.

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